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# A Review and Taxonomy of Choice Architecture Techniques

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ABSTRACT

We present a taxonomy of choice architecture techniques that focus on intervention design, as opposed to the underlying cognitive processes that make an intervention work. We argue that this distinction will facilitate further empirical testing and will assist practitioners in designing interventions. The framework is inductively derived from empirically tested examples of choice architecture and consists of nine techniques targeting decision information, decision structure, and decision assistance. An inter-rater reliability test demonstrates that these techniques can be used in an intersubjectively replicable way to describe sample choice architectures. We conclude by discussing limitations of the framework and key issues concerning the use of the techniques in the development of new choice architectures. Copyright © 2015 John Wiley & Sons, Ltd.

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KEY WORDS behavior change; decision making; choice architecture; nudge; intervention design

The growing field of choice architecture research investigates how the structure and presentation of decision situations influence certain behavioral choices over alternatives (Thaler & Sunstein, 2008). Choice architecture research has gained attention in recent years for its promise to apply insights from behavioral research to areas beyond marketing, such as policy making (Shafir, 2012) or development aid (Banerjee & Duflo, 2011).

Choice architecture emerged when researchers began to take an applied stance on cognitive peculiarities of human decision making drawing upon established judgment and decision making research. The wide focus on deviations in human decision making from the rational choice model ranges from Simon's (1955) bounded rationality proposal and Tversky and Kahneman's (1974) heuristics and biases program to contemporary behavioral economics (Camerer, Loewenstein, & Rabin, 2004) or "applied behavioral science" (Kahneman, 2012, p. ix). Research on choice architectures was triggered by Thaler and Sunstein's (2008) policy-oriented publication *Nudge*, which suggests that researchers should investigate how predictable deviations from rational behavior can be used to "nudge" people into socially desirable directions, for example, improving their health or financial security. A "new branch to the 'prescriptive' part of the field of judgment and decision making" was added (Baron, 2010, p. 224). "Choice architecture" (Thaler, Sunstein, & Balz, 2010) refers to the idea that changes in the decision environment can affect individual decision making and behavior while preserving freedom of choice. The approach alters "people's behavior in a predictable way without forbidding any options or significantly changing their economic incentives" (Thaler & Sunstein, 2008, p. 6). From this, it follows that neither mandates nor classical economic incentives count as choice architecture. In contrast to persuasion, the focus of change is behavior rather than general attitudes or beliefs. Nudges can be understood as a specific type of behavior change technique primarily relying on reflexive cognitive processes (Oliver, 2013) often referred to as cognitive biases or heuristics (Gigerenzer & Todd, 1999; Tversky & Kahneman, 1974). A more general overview of behavior change techniques over and above nudges (including, e.g., mandates and incentives) is presented in the "behavior change technique taxonomy" (Michie et al., 2013).

In recent years, research on how choice architecture interventions can influence individual behavior has been conducted in a growing number of fields, including consumer protection (Shafir, 2006), public health (Blumenthal-Barby & Burroughs, 2012; Cabinet Office, Behavioural Insights Team, 2010), environmental behavior (Cornforth, 2009; Weber, 2012), financial decision making (Thaler & Benartzi, 2004), and development aid (Banerjee & Duflo, 2011). Such research both adapts existing (descriptive) findings about judgment and decision making to the choice architecture paradigm and contributes new insights from research that has been carried out from a (prescriptive) choice architecture point of view (see Baron, 2010 for a similar argument). However, the field still lacks an integrative framework for the development and transfer of successful choice architecture interventions. To provide such a framework, we review the literature, noting both the necessity and the benefits of more clearly distinguishing intervention design from processes, and propose a taxonomy of nine intervention techniques derived through inductive category development from documented cases of choice architecture.

## INTERVENTION DESIGN AND UNDERLYING PROCESSES

Michie, van Stralen, and West (2011) analyzed existing frameworks for behavior change interventions and concluded

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that "none are comprehensive and conceptually coherent" (p. 2). We argue that this applies for the subfield of choice architecture frameworks as well.

As outlined for the broader field of behavior change, specifying the range of available intervention techniques in a systematic and theoretically sound way helps guiding the development of new testable interventions (Abraham & Michie, 2008). Likewise, choice architecture research and the widespread testing in this field would profit from a systematic outline of the techniques available for testing across contexts and domains.

Previous attempts to enumerate techniques of choice architecture have followed two main approaches. The first approach focuses on the underlying (*cognitive*) *processes*, that is, the mental constraints and cognitive biases targeted by an intervention. Datta and Mullainathan (2012) suggest that four basic "mental constraints" serve as a starting point for choice architecture interventions: *self-control*, *attention*, *cognitive capacity*, and *understanding*. Dolan et al. (2012) provide a framework of nine categories that consider psychological phenomena and processes, such as *priming*, *salience*, and *affect*, on which choice architectures can be built. In the second approach, choice architecture techniques are structured according to the kinds of *interventions* used to modify the decision situation. Johnson et al. (2012) suggest as categories of intervention: *reducing the number of alternatives*, *using technology and decision aids*, *setting defaults*, *adjusting the time frames and sequences of choices*, *partitioning options and attributes*, and *designing attributes*. Abrahamse, Steg, Vlek and Rothengatter (2005) single out *commitment*, *goal setting*, *information*, *modeling* (i.e., giving examples of recommended behavior), *feedback*, and *rewards*.

However, these two approaches are often intermixed. Dolan et al. (2012) admit that a possible critique of their taxonomy is the resultant blurring between internal psychological mechanisms, such as affect, and external levers, such as defaults—both of which are part of their taxonomy. Also Datta and Mullainathan (2012) draw upon both approaches when formulating techniques of choice architecture such as "facilitate self-control by employing commitment devices" and "reduce inattention: reminders and implementation intentions". However, such suggestions of direct relationships between mental constraints (e.g., self-control) on the one hand and interventions (e.g., commitment devices) on the other hand cannot claim to be exclusive: reminders, for example, might also strengthen self-control. Moreover, the list of constraints and biases does not explicitly refer to prominent effects such as priming. Similarly, Johnson et al. (2012) describe problems such as *decision inertia*, *alternative overload*, and *long searching processes* and relate them directly to certain interventions, which again suggests an exclusive link between a problem in decision making and a choice architecture technique.

Beyond these two basic approaches, a variety of broad and nonspecific intervention categories are discussed; examples include *encouraging versus discouraging a certain behavior*, *functioning in a mindful versus mindless way* from the perspective of the target person, *being transparent or non-transparent* to the subject (Hansen & Jespersen, 2013; Ly, Mažar, Zhao, & Soman, 2013), *structuring the choice task versus describing the choice options* (Johnson et al., 2012), and *influencing the antecedents versus the consequences of a decision* (Abrahamse et al., 2005). These categories may help to describe a decision situation but provide little guidance in setting up concrete interventions that can be tested empirically.

Summing up, there are several attempts in the field of choice architecture to build a more systematic theoretical framework taking into account the large body of evidence on constraints and biases of human decision making and the resulting opportunities for choice architecture interventions. However, some confusion results from disregarding the factual many-to-many nature of the relationship between intervention techniques (the prescriptive part) and cognitive processes (the explanatory part). One intervention technique might alternatively draw on different cognitive processes, while one such process might underlie different intervention techniques. Consider one of the most prominent examples of choice architecture, default setting: Several processes are discussed to cause default effects, among them loss aversion (Dinner, Johnson, Goldstein, & Liu, 2011), implied endorsement (McKenzie, Liersch, & Finkelstein, 2006), and decision inertia (Johnson & Goldstein, 2003). Conversely, any of these processes can also underlie other intervention techniques besides default setting (e.g., loss aversion might also underlie information translation by reframing gains as losses). Furthermore, if the same process appears in two different techniques, this does not increase similarity from the perspective of the choice architect who might, for example, only be able to intervene on the level of information framing but not on the level of default setting. As a consequence, attempts to more stringently systematize the field need to opt for *either* techniques (cf. Johnson et al., 2012) *or* processes (cf. Vlaev & Dolan, 2015) as the basic categorization logic.

We provide a suggestion of how to use intervention techniques as the structuring principle. We view this alternative as particularly suited for facilitating the development of new, testable choice architecture interventions.

## INTEGRATION WITH AN UNDERLYING MODEL OF BEHAVIOR

The development process for choice architectures can be specified as a systematic approach of sequenced steps as, for example, suggested by Datta and Mullainathan (2012) for development aid policy. In the following, we propose a more general and revised version building on the "behaviour change wheel" model by Michie et al. (2011):

Step 1. Define behavioral problem and target behavior;
Step 2. Analyze applicability of choice architecture framework;
Step 3. Check for behavioral bottlenecks; and
Step 4. Build hypotheses on promising choice architecture interventions



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Initially, any choice architecture intervention (and any behavior change attempt in general) requires a definition of the target behavior. Second, the kind of behavior change approach applicable to the behavioral context should be determined. This includes checking whether choice architecture is applicable at all. If people strongly oppose a behavior or are forced not to display it because of external factors, choice architecture does not appear suited. Alternative measures of behavior change (such as education, bans or mandates, or economic incentives) might be more promising than choice architecture. However, choice architecture might be pertinent for pragmatic reasons if alternative measures are not available, too expensive, or not admissible.

Third, the reasons why people fail to display the target behavior should be analyzed. The crucial question is whether they are to be found in the psychology of human decision making. This analysis has been termed the check for a "behavioral bottleneck" (Datta & Mullainathan, 2012, p. 15). It yields hypotheses on why the target behavior is not displayed, opening up a path towards choosing potentially effective intervention techniques. To do so, we suggest focusing on the central determinants of behavior in the "behaviour change wheel" (Michie et al., 2011): capability, opportunity, and motivation, and to identify corresponding biases or constraints such as limits in self-control, attention, cognitive capacity, and understanding (Datta & Mullainathan, 2012). Possible results of this analysis could be that *capability* is constrained by a lack of understandable information (e.g., complicated information on the costs of insurance alternatives); the physical setup might negatively impact *opportunity* (e.g., inaccessible healthy food options), or self-control deficits might undermine the translation of *motivation* into behavior (e.g., in smoking cessation attempts).

Because of the complex influence of multiple variables, a general description of the analysis for behavioral bottlenecks necessarily remains vague. Thus, there can be no straightforward how-to for connecting the results of the bottleneck analysis with distinct choice architecture intervention techniques as a fourth step. Nonetheless, knowing about pertinent biases or constraints that prevent a target behavior while step-by-step checking each of the intervention techniques from the presented taxonomy, provides a more systematic way of developing choice architectures than currently available. In the following section, we first describe the development of our framework of choice architecture techniques and then explain the techniques in more detail.

## A TAXONOMY OF CHOICE ARCHITECTURE TECHNIQUES

### Taxonomy development

While a large range of techniques for behavior change, including choice architecture, has been described, particularly in health and social psychology, they tend to be presented "as practical tools without reference to their evidence base" (Michie, Johnston, Francis, Hardeman, & Eccles, 2008, p. 665). To remedy this fact, we based category development on a sample of 127 documented examples of empirically tested choice architecture interventions. These were collected from academic publications and practitioner reports provided by organizations such as the British Cabinet Office's *Behavioural Insights Team* and MIT's *Abdul Latif Jameel Poverty Action Lab*. Examples mentioned in blogs or non-academic publications were also included after the original publication was checked for academic soundness.

The aim of any taxonomy or classification scheme is to "categorize phenomena into mutually exclusive and exhaustive sets" (Doty & Glick, 1994, p. 232) so that those sets describe the greatest possible share of the respective phenomena. To adopt a traceable approach to develop categories of choice architecture techniques meeting these requirements, we used inductive category development (Chenail, 2008; Corbin & Strauss, 2008; Mayring, 2010). Just as quantitative data reduction techniques (e.g., principal component analysis and cluster analysis), inductive category development aims at providing a limited number of categories suited to describe a given data set. The method is used to identify thematic relationships between units and to build categories through a recursive process of analyzing units and accordingly creating new or adapting existing categories (for a similar application, see Abraham & Michie, 2008; Michie et al., 2011). Ultimately, saturation in category development is expected because—if an adequate set of categories could be developed—no further categories need to be defined, but further units can be subsumed under the previously defined categories (Corbin & Strauss, 2008). In order to define the units for analysis, we first compiled a unified description for all 127 interventions in our database, specifying (a) the actors, (b) the desired behavior, (c) a description of the intervention, and (d) the effect of the intervention on behavior. Processing each of these units, we then developed and recursively refined a set of descriptive categories of types of interventions suited to subsume all studied interventions. For each newly processed intervention, we checked whether it could be subsumed under the definition of one of the previously created categories, required further specification of the category's definition, or required the creation of an additional category. To maximize intersubjectivity during category development, all three authors discussed the sample cases in an iterative process. The sample database was large enough to continue category development to the point of saturation where all further sample interventions could be subsumed under existing categories, resulting in a stable classification scheme (cf. Eisenhardt, 1989). We thus found that heterogeneous choice architecture *interventions* can be described using a limited number of common choice architecture *techniques*.

The resulting taxonomy was then subjected to an inter-rater reliability test to quantify the degree of how well the taxonomy enables independent others to subsume given choice architecture interventions under the pre-defined categories. Four coders (two female and two male) categorized a random subset of 55 of the 127 choice architecture examples (see Supporting Information for a table containing descriptions and references of all 55 coded interventions). Prior to the categorization, all coders received written definitions of techniques and examples, which they were free to



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use during coding. After practicing with 20 examples, the coders were presented with the target cases in a random, identical order. Complete inter-rater agreement was achieved in 47% of cases; at least three coders agreed in 73% of cases, and only 5% (three cases) were rated entirely different. Given that complete agreement by chance is extremely unlikely (0.14% per case), the high agreement rates already indicate inter-rater reliability. Following Hallgren's (2012) recommendations for fully crossed designs with nominal data, Cohen's kappa was computed for each pair of coders and then averaged to provide a single index (Light, 1971). Pairwise kappas lie between $\kappa = .56$ and $\kappa = .65$; all $p < .001$. The overall mean kappa of $\kappa = .60$ ($SD = .03$) provides a conservative estimate of inter-rater reliability and indicates moderate to substantial agreement (Landis & Koch, 1977).

## Structure of the taxonomy

We suggest three basic categories of choice architecture intervention techniques: *decision information*, *decision structure*, and *decision assistance* (Table 1). These categories reflect different streams in the judgment and decision making literature.

### *Decision information*

How people process available information has been a core topic in behavioral decision research since Simon's (1955) bounded rationality claim. Scholars have pointed to the relevance of "perceptual processes of problem representation, formulation, or framing" (Payne, Bettman, & Johnson, 1992, p. 111) to decision making. Accordingly, this category covers choice architecture techniques that target the presentation of decision-relevant information without altering the options themselves, for example, by (re)arranging existing information or changing its presentation (how and by whom).

Table 1. Choice architecture categories and techniques

| Category | Technique |
|---|---|
| A. Decision information | A 1 Translate information<br>*Includes*: *reframe*, *simplify* |
| | A 2 Make information visible<br>*Includes*: *make own behavior visible (feedback)*, *make external information visible* |
| | A 3 Provide social reference point<br>*Includes*: *refer to descriptive norm*, *refer to opinion leader* |
| B. Decision structure | B 1 Change choice defaults<br>*Includes*: *set no-action default*, *use prompted choice* |
| | B 2 Change option-related effort<br>*Includes*: *increase/decrease physical/financial effort* |
| | B 3 Change range or composition of options<br>*Includes*: *change categories*, *change grouping of options* |
| | B 4 Change option consequences<br>*Includes*: *connect decision to benefit/cost*, *change social consequences of the decision* |
| C. Decision assistance | C 1 Provide reminders |
| | C 2 Facilitate commitment<br>*Includes*: *support self-commitment/public commitment* |

### *Decision structure*

Many decision making models assume that decision makers compare attributes or alternatives. Because maximizing strategies often prove impossible or maladaptive (Schwartz et al., 2002), alternative strategies have been suggested (e.g., weighted additive, satisficing, and lexicographic; see Payne, Bettman, & Johnson, 1988 for an overview). Accordingly, choice architects can alter the decision structure, that is, the arrangement of options and the decision making format, by modifying the available options in the decision situation, including their range or composition, the default option, or the effort required for selecting an option and the consequences of selecting it.

### *Decision assistance*

A third stream of decision research has focused on self-regulation and "self-regulation failures" (Baumeister & Heatherton, 1996), that is, problems of bridging the intention–behavior gap (Sheeran, 2002). Accordingly, choice architects can use techniques to provide decision makers with further assistance, which aims at helping them to follow through with their intentions. For example, choice architects can foster deliberate commitment or take measures to remind people of preferred behavioral options.

Each basic category contains a number of techniques. In the following sections, we will describe the techniques and cite examples of their effective application from empirical studies. The examples are structured according to subtypes, but the subtypes given do not constitute an exhaustive list. We will also refer to underlying cognitive processes which—as illustrated earlier—are not exclusively linked to a technique.

## Decision information

Decision makers usually base their decisions on available information, be it factual or social. The way this information is presented influences subsequent decisions, and thus, changing the presentation of information can be seen as part of the choice architect's role (Thaler & Sunstein, 2008).

### *Translate information (A1)*

Choice architects can translate existing, decision-relevant information for the decision maker by changing the format or presentation of information but not the content.

*Reframe.* One way to change the format of existing information is to reframe it by "shifting the perspectives of decision makers in ways that change their subjective evaluations of choice options" (Weber, 2012, p. 387). Following classical definitions, framing effects occur when the same (equivalent) information presented in different ways





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leads to systematically different decisions (Sher & McKenzie, 2011). According to this definition, translating information by reframing it includes formally, logically, and mathematically equivalent information presentation as illustrated by the following examples. Framing the call for blood donations as death-preventing rather than as life-saving raised the amount of blood donations in one experiment (Chou & Murnighan, 2013). Another study using this technique aimed at increasing teacher performance. By paying end-of-the-year bonuses for teachers in advance, bonuses were presented as conditionally awarded money that had to be paid back at the end of the year in case of poor performance (Fryer, Levitt, List, & Sadoff, 2012). But harnessing loss aversion is only one process that the reframing technique can draw on. Because the framing of probability and quantity estimates affects their communicative content (Halberg & Teigen, 2009), a reframing intervention can aim at influencing patients' decisions when confronted with health statistics (Gigerenzer, Gaissmaier, Kurz-Milcke, Schwartz, & Woloshin, 2007). Whereas strict definitions of framing require equivalence, the term is often also used in a more loose sense including "linguistic redescriptions of the same decision problem" (Krüger, Vogel, & Wänke, p. 13). For choice architects, it follows that variants of the reframe technique may not be equivalent as long as they target the same behavior and refer to the same decision problem. Reframing hygiene measures in a hospital to emphasize the patient's rather than the doctor's health is such an example of reframing decision-relevant information to target doctors' overconfidence in their own resilience (Grant & Hofmann, 2011). Clearly, the health frames are not logically equivalent (because doctors' health and patients' health are not the same) and do not suffice the criteria of a strict framing definition. However, the same target behavior (hand hygiene) is described with the same information (hygiene prevents disease) by varying the framing of its consequences (me versus others) and thus suffices a loose framing definition.

To count as a reframing, non-equivalent frames must not reveal new information as, for example, by making it visible (cf. Technique A2) but rather shift the focus of decision makers by presenting existing information differently. This is exactly what equivalent frames also do.

*Simplify.* An alternative way of making existing information more helpful to decision makers is simplification, that is, reducing the burden of cognitive effort necessary to process the information available and increase its usefulness in the decision making process. Simplification can, for instance, adjust for constraints in cognitive capacity and understanding. This technique can be easily implemented and substantially improve decision making because in many situations, relevant information is theoretically available but practically underused. Simplified information has been shown to increase college enrolment rates and financial aid application rates (Bettinger, Long, Oreopoulos, & Sanbonmatsu, 2009). Micro-entrepreneurs with little knowledge of accounting or economic decision making also profit from simplification, such as the provision of rules of thumb (Drexler, Fischer, & Schoar, 2010). Further simplifications concern numerical information, for example, on fuel efficiency (Larrick & Soll, 2008; Soll, Keeney, & Larrick, 2013). Generally, simplification facilitates processing existing information in a given decision situation and refers only to available information that is simplified (e.g., by translating it into plain language or understandable numerical formats). By contrast, reducing the number of options available to facilitate processing does not count as a simplification because it changes the decision structure and goes beyond information translation.

*Make information visible (A2)*

Frequently, the information necessary for making a decision or for changing behavior is not apparent or readily available. For example, the daily amount of calories we consume can be made available for decision making by simple measurement and disclosure. Many current approaches to public policy strive to provide easier access to information that is normally invisible, such as annual credit card statements informing consumers how they have used their credit card or energy performance certificates containing information about house insulation (cf. Cabinet Office, Behavioural Insights Team, 2011b, chapter 1).

*Make own behavior visible (feedback).* Feedback can have a powerful influence on behavior (Hattie & Timperley, 2007). In particular, behavioral consequences are often invisible in situations where feedback is infrequent or temporally and spatially disconnected from decision making. Such a lack of feedback manifests in both common behaviors (people taking a shower lack feedback on their consumption of water and energy) as well as infrequent situations (the annual utility bill aggregates a multitude of past energy consumption behaviors). More direct feedback on behavior can have various forms, including smart electricity meters displaying energy consumption (Jessoe & Rapson, 2014) and segmentation cues, such as including a red potato chip for every five chips, allowing an individual to track the amount of food already consumed (Geier, Wansink, & Rozin, 2012). In many cases, providing feedback about one's own behavior counteracts the constraints in attention and processing capacity that make this information inaccessible in daily life. Tools and devices that provide feedback (such as pedometers for counting steps) are becoming increasingly popular as part of a recent trend in self-optimization (Lubans, Morgan, & Tudor-Locke, 2009).

*Make external information visible.* Apart from the consequences of one's own behavior, much external decision-relevant information is also frequently unavailable, and making this information visible can empower decision makers. For example, when information about restaurant hygiene was bundled and conveniently displayed at the front door with a colored label, people could more easily choose to avoid unsanitary restaurants and the incidence of foodborne disease was reduced (Simon et al., 2005). Sponsor-a-child programs work similarly by ensuring the visibility of the donation's purpose (Small & Loewenstein, 2003). The visibility of relevant information is already a key tenant of



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consumer protection (e.g., food nutrition labeling). The American government website www.data.gov provides another example of a large-scale attempt to facilitate access to external information. In contrast to feedback, which is, in principle, available but requires effort to obtain, external information as mentioned in the examples earlier usually remains inaccessible if it is not made visible through choice architecture.

### *Provide social reference point (A3)*

People neither make decisions nor behave in total isolation but in a social and cultural environment. Within this environment, the behavior of others influences the behavior of the individual and serves as a social reference point (Cialdini & Goldstein, 2004). The behavior of other people can appear in the form of group behavior or behavior of specific persons valued for special reasons, such as knowledge, fame, group membership, or a specific function (teacher, parent, or role model). Providing such social reference points is therefore a choice architecture technique that encompasses Allport's (1968) observation that "the thought, feeling and behavior of individuals are influenced by the actual, imagined, or implied presence of others" (p. 3).

*Refer to descriptive norm.* Descriptive social norms depict the observable behavior of other people, that is, what others actually do, in contrast to injunctive norms describing what one *should* do (Bicchieri & Xiao, 2009; Cialdini, Reno, & Kallgren, 1990). Choice architecture interventions, which refer decision makers to pertinent descriptive norms, might draw on a variety of cognitive processes. According to Cialdini and Goldstein (2004), what drives people to follow norms is situational ambiguity and behavioral uncertainty combined with the goals to behave appropriately, receive approval, and maintain a positive self-concept. This manifests, for example, in experiments on the bystander effect, in which people ignored smoke in a room if other people in the room ignored it (Latane & Darley, 1968). The need to belong (cf. social identity theory, Tajfel & Turner, 1986) and the fear of ostracism (cf. Cialdini & Goldstein, 2004) are further drivers for aligning one's behavior with others.

Empirical evidence about the efficiency of social norms comes from various fields. People informed about the energy consumption or recycling behavior of their neighbors adjusted their own behavior as a consequence (Allcott & Mullainathan, 2010; Cotterill, Moseley, & Richardson, 2012; Dolan & Metcalfe, 2013; Schultz, Nolan, Cialdini, Goldstein, & Griskevicius, 2007). In a classic study, the reuse of towels in a hotel room depended on the information people were given stating whether other hotel guests had used their towels more than once (Goldstein, Cialdini, & Griskevicius, 2008). Aside from environmental behavior, social norms have been shown to help lessen student substance abuse by pointing out that the majority of students do not drink to excess (DeJong et al., 2006; Perkins, 2003). Contributions to charity (Shang & Croson, 2009), environmental protection behaviors (Cialdini, 2003), and voting in elections (Bond et al., 2012; Gerber & Rogers, 2009) are also influenced by social norms. As all techniques, referring to the descriptive norms might also work contrary to the intention of the choice architect. Complaints about low voter turnout or teacher absence in developing countries can worsen such problems by the same process that can be used to improve them: the contagious impact of others' behavior (Chaudhury, Hammer, Kremer, Muralidharan, & Rogers, 2006).

*Refer to opinion leader.* As known from dual processing models (Chaiken, 1980; Petty & Cacioppo, 1986), the source of information matters in addition to its contents. Highly valued, respected messengers (e.g., experts or role models) can influence opinions and behavior. From advertising to communicating health topics or technological innovation, opinion leaders are used as information disseminators to improve the impact of campaigning (Leonard-Barton, 1985; for a review, see Valente & Pumpuang, 2007). In development aid, changing the behavior of opinion leaders has been shown to improve the acceptance of safer and healthier non-traditional cook stoves (Miller & Mobarak, 2013). This technique makes use of peripheral processing (Petty & Cacioppo, 1986). With increasingly far-reaching social networks and heightened complexity in many domains, opinion leaders appear to be powerful social reference points, and thus, this intervention technique can be a useful tool for choice architects.

## **Decision structure**

A choice architect will not always be able to influence information presentation and as a consequence may be unable to use one of the techniques described earlier. In such cases, the choice architect can revert to techniques addressing the decision structure, that is, the arrangement of options and the decision making format, which includes setting defaults, rearranging the composition of options, and changing option-related efforts or consequences.

### *Change choice defaults (B1)*

Defaults are pre-selected options that leave decision makers the freedom to actively select a different option. Research has shown that in many situations people accept the default. This effect is present in both minor decisions such as online privacy settings (Johnson, Bellman, & Lohse, 2002) and more important decisions concerning pension savings (Thaler & Benartzi, 2004), end-of-life care (Halpern et al., 2013), and organ donation (Johnson & Goldstein, 2003). The effect of defaults on behavior is caused by a number of different processes, including effort reduction, implied endorsement, and the unwillingness to give up the status quo when it is understood as an endowment (Dinner et al., 2011; McKenzie et al., 2006).

*Set no-action default.* No-action defaults "refer to what happens in the absence of choice" (Dinner et al., 2011, p. 332) and can range from universal mass defaults to custom-made personalized defaults based on past behavior or past choices



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(Goldstein, Johnson, Herrmann, & Heitmann, 2008); Sunstein, 2013). Clever defaults can also help bridge the gap between intentions and behavior, such as in the choice of green electricity, which is favored by far more people than those who actually opt for it. When green energy was the default in a field experiment, more people used green energy compared with the traditional gray energy that was offered as the default in most communities (Pichert & Katsikopoulos, 2008; Sunstein & Reisch, 2013). Other fields in which setting defaults is used as a choice architecture intervention are financial behavior (Beshears, Choi, Laibson, & Madrian, 2009), medical decision making (Ansher et al., 2014), and health care (Halpern, Ubel, & Asch, 2007). Even room temperature in offices depended on the default in a field experiment, although not when the default setting was too extreme (Brown, Johnstone, Haščič, Vong, & Barascud, 2013). The size of a unit (e.g., a cup, a plate, and a suitcase) also serves as a default. In restaurants, it may often be plate size rather than hunger that determines how much people eat (Wansink, 2006), as plate size is often taken for granted as an indicator of the appropriate amount of food (Geier, Rozin, & Doros, 2006; Kallbekken & Sælen, 2013). As illustrated by these very heterogeneous examples, default setting can be applied successfully to promote behavior change in many different fields. Successfully using defaults in choice architecture is easier with homogeneous target groups. For controversial decisions or for very heterogeneous target groups, determining the optimal default can prove difficult (Choi, Laibson, Madrian, & Metrick, 2003).

*Use prompted choice.* For heterogeneous target groups, prompted choice, that is, forcing people to actively decide without a pre-existing default, can be more suitable than defaults. Prompted choice avoids the status quo bias or default effects because of inertia or assumed recommendations. However, enforcing active choice and deliberation may give cognitive biases more influence because of greater reflection (Amir & Lobel, 2008). Thus, one bias (status quo bias) might just be replaced by another bias (e.g., choice aversion). As discussed by Sunstein (2014), further complexities might arise through prompted choice especially when people actively "choose not to choose". With the power and limits of defaults in mind, it must be a case by case decision of the choice architect whether to actively set defaults or make a decision mandatory. In any case, knowledgeable choice architects can avoid disadvantageous defaults, such as one-sided printing preselected at most printers (Egebark & Ekström, 2013).

### *Change option-related effort (B2)*

The effort involved in choosing an option plays an important role in determining which of several options people choose. The higher the effort for choosing an option, the more this effort acts as a barrier. However, to qualify as choice architecture and differ from standard economic transaction costs, the change in effort should only be marginal, which excludes imposing unsurmountable (economic or behavioral) costs on the decision maker (which would clearly not be in line with the metaphor of just giving a slight "nudge," either). What exactly counts as a marginal effort is difficult to quantify and differs between situations and people, but relatively little extra effort should not justify behavior change on rational grounds. Changing physical or financial effort modifies the decision structure. Note that choice architects can also modify the cognitive effort required; however, this alters the way decision-relevant information is presented and is thus covered in the section on decision information.

*Increase/decrease physical effort.* The physical effort necessary to realize a behavior is determined by a variety of factors, such as the accessibility or distance of a target object. The little extra effort introduced by a choice architect should not justify behavior change according to a rational cost–benefit analysis or else efforts would be standard economic transaction costs. Still, these marginal physical facilitations or barriers can lead to significant behavior change. Making an apple easier to grasp than less healthy alternatives is discussed as an example for promoting healthy choices in a situation where hunger or appetite depletes self-control, leading to regrettable decisions. More generally, facilitating access to healthy food options and increasing the effort needed to choose the unhealthy option by "changing places" (Ashe, Graff, & Spector, 2011) can support people in their intention to eat healthier (Wansink, 2004). In development aid, decreasing the effort for farmers to obtain fertilizer simply by delivering it to them helped raise fertilizer use and thus increased crop yield (Duflo, Kremer, & Robinson, 2011). The effort to clear out the attic entailed by opting for installing an energy-saving insulation is another example for a physical effort that functions as a "hassle factor" (Caird, Roy, & Herring, 2008) for people to use more efficient insulation.

*Increase/decrease financial effort.* We suggest the term "financial effort" to designate issues paralleling the concept of physical effort in decisions concerning financial transactions. Independent of the actual amount in question, people perceive the realization of financial transactions as more or less costly. Choice architects can intervene to change the factors determining this perception of financial effort. Note that the economic fundamentals of an option such as the final price remain unchanged or else the technique would not differ from classical benefits. Examples include postponing costs to the future without changing actual final costs. This intervention was suggested by the British government in their "Green Deal": allowing citizens to pay for more energy efficient appliances or technologies using the savings generated by these should reduce the barrier of adopting green technology, which pays off in the long run but poses high costs immediately (Cabinet Office, Behavioural Insights Team, 2011a). The otherwise discouraging financial effort tied to an investment that ultimately pays off is thus spread out over a larger period and perceived as smaller. Human tendencies to discount the future (costs and benefits) more than the present and limited self-control lead to different perceptions of the effort connected to a financial transaction depending on its timing (Loewenstein & Prelec, 1992; O'Donoghue & Rabin,



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1999). Correspondingly, people save more money if they can opt for raising the amount saved in the future as opposed to increasing it on the spot (Thaler & Benartzi, 2004). The same applies to donations: when people were asked to increase their donations starting in the future ("Give more tomorrow"), the increase in donations was 32% higher than in a control group asked to give more immediately (Breman, 2011). Furthermore, small but frequent donations are not perceived to be as costly as single large donations. Choice architects can make use of this so-called "peanuts effect" (Weber & Chapman, 2005) by shaping the perception of effort connected to a financial decision.

### *Change range or composition of options (B3)*

Choice architects can not only change *how* something is presented but also *what* is presented to influence the relative attractiveness of choice options (Chang & Liu, 2008). Assuming that fixed preferences do not always guide behavior entails that many preferences are constructed in the decision situation and thus depend heavily on the alternatives offered (Ariely, Loewenstein, & Prelec, 2003; Slovic, 1995). For example, decoy alternatives (Huber, Payne, & Puto, 1982), that is, the introduction of additional options, are used strategically to influence consumer decisions (Heath & Chatterjee, 1995).

*Change categories/grouping.* Because resources like money or time are limited for most individuals, they have to be allocated across different goals following some rationale. This necessity is a gateway for cognitive biases, among them the diversification bias, that is, the tendency to divide resources by the amount of available categories and allocating them evenly (Fox, Ratner, & Lieb, 2005). A related bias is mental accounting (Thaler, 1999), that is, treating a fungible resource like money as non-fungible between purposes (borrowing money to avoid taking money from a savings account). Studies have also found that a higher amount of money is spent when it appears in small units ($1 bills) than larger units ($20 bills) (labeled "denomination effect" by Raghubir and Srivastava, 2009). Allocation biases, such as the diversification bias, variety seeking, mental accounting, and the denomination effect, have a large influence on how people allocate available resources in decision situations. A choice architect can thus tactically arrange the respective categories or allocation alternatives, such as segregating healthy options into more diverse categories (Kahn & Wansink, 2004), partitioning safety and style attributes of a car differently (Martin & Norton, 2009), or presenting more fine-grained, segregated charity purposes instead of presenting an overall goal (Fox et al., 2005).

### *Change option consequences (B4)*

Choice architects can also modify the consequences of decision options. This appears similar to incentivizing or disincentivizing particular behaviors. However, in contrast to the classical rational choice cost–benefit paradigm (according to which considerable (monetary) reasons for or against a behavior are provided), choice architecture provides "micro-incentives," that is, changes of the consequences of decision options that are insignificant from a rational choice perspective (e.g., providing an improbable chance of winning a lottery or requiring middle class shoppers to pay five cents for a shopping bag, cf. Homonoff, 2012). If this kind of intervention is effective, the choice behavior is affected by processes aside from rational cost–benefit analyses. From an intervention perspective, one can distinguish the manipulation of small, individual monetary benefits/costs from the social consequences of a choice option.

*Connect decision to benefit or cost.* Independent of the effort directly entailed by realizing a behavior (cf. earlier), the behavior might trigger additional costs or benefits. Connecting a desired behavior to a small benefit or an undesired behavior to a small cost changes the probability of occurrence, even if the respective benefit or cost is too small to "significantly [change the] economic incentives" (Thaler & Sunstein, 2008, p. 6). While we acknowledge that it is difficult to draw a hard line separating "significant" from "insignificant" incentives, there are some clear examples. In one study, introducing a five-cent tax for shopping bags substantially reduced bag use in US middle class supermarkets (Homonoff, 2012), but five cents is not a significant monetary incentive to middle class shoppers. In another study, offering participation in a lottery for each day that people took their medication properly generated an insignificant chance for a small payoff but did increase the regularity of taking medication (Volpp et al., 2008). In line with Thaler and Sunstein (2008), we posit that interventions involving such "micro-incentives" as a five-cent tax for shopping bags are different from classical incentive schemes such as 35% price cuts. However, it is not yet understood precisely how micro-incentives work. Lotteries might harness the tendency to overweigh small probabilities, whereas a micro-tax or the prospect of missing out on a small benefit already mentally booked as property might harness loss aversion. Note that choice architecture cost/benefit interventions serve as *additional* motivators for people, not as persuasive arguments as in advertising.

*Change social consequences.* The range of consequences that a choice option might have for an individual extends beyond (small) monetary costs and benefits. For example, consequences might also concern the social integrity of the individual. Choice architects can connect the choice of a specific option with the consequence to be regarded more positively or more negatively by others. This might draw, for example, on image motivation as the desire to be liked and well regarded by others, a concept that has been suggested to be "a driver in prosocial behavior" (Ariely, Bracha, & Meier, 2009). Likewise, competitive altruism (Barclay & Willer, 2007; van Vugt, Roberts, & Hardy, 2007) and costly signaling theory (Glazer & Konrad, 1996; Griskevicius et al., 2007) have been proposed to explain altruistic decisions through the socio-communicative messages associated with altruistic behavior and the desire for social approval. In addition to demonstrating socially desirable behavior, altruistic decisions signal the ability to spend money, time, or effort



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on these decisions without suffering from disadvantages. In one study, activating status motives made people choose environmentally friendly products more often in a public situation where positive self-presentation through choice–behavior was possible (Griskevicius, Tybur, & van den Bergh, 2010). Choice architects can create such self-serving presentation possibilities for a desirable behavior. As an example for connecting a behavioral alternative to negative social consequences, in the Grameen model of microcredit, a loan is given to each member of a group only after the previous loan receiver has returned the loan. Not repaying is thus connected to the consequence of peer pressure by the others waiting for their loan (Auwal, 1996).

**Decision assistance**

Besides working on information and options, choice architects can provide decision makers with further assistance to help them follow through with their intentions. To do so, choice architects can remind individuals of the preferred alternative in the decision situation and foster deliberate commitment to beneficial actions.

*Provide reminders (C1)*

Within the daily flood of information and new stimuli, information that is salient and easy to access has a higher chance of guiding behavior and decisions. Oftentimes, however, important information is not salient, and thus, because of limits in attention and cognitive capacity, it is not considered. Choice architects can intervene by providing positive reminders that heighten the salience of a desired option, for example, reminding people of socially desirable concepts such as voting (Greenwald, Carnot, Beach, & Young, 1987) or honesty (Shu, Mazar, Gino, Ariely, & Bazerman, 2012). More specifically, reminding bank clients via text messages or letters of saving more increased savings in a field study, especially if reminders highlighted particular savings goals (Karlan, McConnell, Mullainathan, & Zinman, 2010). In addition to providing positive reminders, choice architects can also intervene to oppress cues, which remind people of an undesired option. Choice architects can diminish the salience of undesired options or external cues that hint towards them by, for example, putting unhealthy food in non-see-through containers (Wansink, Just, & Payne, 2009), or positioning unhealthy food options in the middle of the menu instead of at the beginning or at the end where they would have primacy or recency advantages of being chosen (Dayan & Bar-Hillel, 2011; Li & Epley, 2009).

Providing reminders should not be confused with the *make information visible* technique, which refers to making previously unknown or inaccessible information accessible to decision makers; providing reminders only changes the position of familiar stimuli by moving them into or out of the attention focus.

*Facilitate commitment (C2)*

Private or public commitment towards certain behaviors makes individuals more likely to follow through because it counteracts self-control problems. Deviations lead to cognitive dissonance or the need to justify the deviation in front of others. Facilitating commitment is thus a way to help people to overcome constrained self-control and bridge the intention–behavior gap.

*Support self-commitment.* As research on self-commitment has shown, people understand their own deficits in self-control such as temptation or procrastination and try to work against them by self-imposing deadlines (Ariely & Wertenbroch, 2002). Commitment devices, that is, arrangements "with the aim of helping fulfil a plan" (Bryan, Karlan, & Nelson, 2010, p. 671), such as websites offering to formalize a commitment and set up a penalty for deviance (e.g., www.stickk.com), are choice architecture interventions tailored to support willpower when it reaches its limits. Other suggestions include addicted gamblers putting their name on a ban list (Thaler & Sunstein, 2008), browser applications blocking Internet access for specific times, or depositing money to foster smoking cessation (Giné, Karlan, & Zinman, 2010). Among the processes harnessed by many of these self-commitments is cognitive dissonance that arises when goals (commitments) and action (behavior and decisions) are inconsistent.

*Support public commitment.* In a similar vein, commitments can be made in front of other people, thus introducing another "supervisor" in addition to the self. Publicly committing to a behavior (working out) or to refrain from a behavior (smoking) creates external pressure and possibly negative consequences in case of breaking the commitment (like face-management problems or a heightened need to justify one's behavior). Effects of public commitment have been found in studies on recycling behavior (Cotterill, John, Liu, & Nomura, 2009; DeLeon & Fuqua, 1995), weight loss (Nyer & Dellande, 2010), or formalized agreements between parents and schools (Evans, Hall, & Wreford, 2008). The degree of formality of public commitment interventions can vary from simple public announcements or pledges up to formal written agreements, but in all cases, the commitment is voluntary, preserving freedom of choice as in any choice architecture technique. The choice architect's task is to create and support public commitment possibilities.

Note that *public commitment* and *changing the social consequences* of a decision may work because of the same process (social pressure) but still represent separate interventions, as they are of different intervention types: connecting a choice to a social consequence (e.g., publicly appearing as an altruist/egoist) is a different intervention than providing an opportunity for public commitment. Public commitment presumably does not work exclusively because of anticipated social pressure but also because of further processes like preference for consistency (Cialdini, Trost, & Newsom, 1995). This illustrates our claim that, even if we separate interventions from processes, process overlap between intervention techniques remains possible. However, this overlap is exactly what allows the choice architect to design interventions: while she might not be able to install a commitment device, she might be able to connect a decision option with



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a positive social consequence. Thus, the separation of intervention techniques that work partially because of the same processes ensures the greatest flexibility for the choice architect to design intervention strategies.

## DISCUSSION

We have proposed a theoretical framework for choice architecture consisting of nine intervention techniques inductively derived from an analysis of documented empirical examples of choice architecture. The techniques target the provision of *decision information*, changes in the *decision structure*, and measures of *decision assistance*. These techniques structure options for designing choice architecture *interventions* as opposed to the underlying processes (cognitive biases or mental constraints) and thus belong to the prescriptive branch of the judgment and decision making tree (cf. Baron, 2010). A potential benefit of a systematic framework of techniques summarizing concrete cases is its power to sharpen the concept of choice architecture in a bottom–up approach.

The nine techniques are conceived of as ideal types, which can be found in different combinations in real-world situations. Furthermore, as we have pointed out, differentiating between the techniques is ultimately an issue of definition (cf. *reminders* versus *make visible* or *public commitment* versus *social consequences*). While the chosen method of inductive category development facilitates the development of a testable taxonomy, both throughout category development (developing definitions) and in using the categories (applying the definitions), subjective decisions are unavoidable. Hence, raters may disagree on the correct classification of a specific intervention, in particular, if raters differ by educational background, training, or experience. On the other hand, a quantitative method to reduce data and detect latent categories like multidimensional scaling, even if requiring less interpretation, would hardly lead to a sufficiently fine-grained solution. This holds true, in particular, for categorizing choice architecture interventions that cannot easily be transformed into numerical data but are available as text data. Qualitative differences between such interventions are thus better captured using a qualitative method. It is important to take measures to maximize intersubjectivity, both during category development (all cases were discussed by at least three researchers) and after finalizing category development. Therefore, inter-rater agreement in applying the categories was tested with four independent raters. The resulting kappa coefficient reflects the expectable amount of subjectivity involved while still allowing to conclude that the categories have proven intersubjectively replicable.

Given the growth of new evidence from the heterogeneous field of choice architecture, we expect an inductively generated framework such as the one we propose to further develop in the future. As this is a cumulative process, additions are likely (Michie et al., 2008), but we assume that additions will primarily concern the subtypes, which illustrate the techniques rather than representing an exhaustive list of subcategories.

We agree with Lunn (2012), who concludes in a review on choice architecture in policy making that the field is too complex to suggest a straightforward toolkit for generalists. Instead of a toolkit, we suggest that choice architects use the nine techniques to determine which one could be applicable to a specific challenge. By synthesizing evidence and providing a method to structure and evaluate interventions, taxonomies support the transfer of and access to successful interventions (Michie, Jochelson, Markham, & Bridle, 2009). This has been demonstrated domain-specifically, for example, in public health (Michie et al., 2012) as well as across domains (Michie et al., 2013). Similarly, the presented framework is expected to facilitate the identification of potentials (generalizations) and pitfalls. Specific life circumstances (e.g., poverty, Mani, Mullainathan, Shafir, & Zhao, 2013) or interpersonal differences (Beshears, Choi, Laibson, & Madrian, 2010) might impede transfer of interventions across situations. Thus, empirical testing remains crucial for clarifying what works.

To date, empirical evidence on the limitations and success conditions of choice architecture remains scarce (see Heilmann, 2013 for a methodological perspective on success conditions). Setting aside those situations in which choice architecture is not the most promising approach, it is clear that where it works, it likely provides a less costly way to change behavior than alternative measures. If we have a clearly defined target behavior and understand how cognitive biases or constraints affect the behavioral target-actual gap, then the proposed framework can contribute to designing promising testable choice architecture interventions.

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